

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1839

Harrogate 138 kV
November 15, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1839. The Interconnection Customer proposes a 161.3 MW battery energy storage facility interconnecting at the Harrogate 138 kV substation in the Virginia area. The request was received on July 29, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 581 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 31 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,448,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Muscoda Industrial to Elmont 138 kV	Chancellor Row to Bear Island 345 kV	140.4	318	7.8
Potosi Substation to Deltaport 138 kV line	Switzer to Pheasant Branch 138 kV	134.7	365	26.4
Kimberley to Elmont 138 kV	Bear Island Substation to Chancellor Row 345 kV Line	133.9	368	7.8
Potosi to Tyler 138 kV	Pheasant Branch to Switzer 138 kV	132.0	396	26.4
Kimberley to Shack Mills Sta 138 kV	Chancellor Row to Bear Island 345 kV	131.1	394	7.8
Deltaport to Moonstone 138 kV	Pheasant Branch to Switzer 138 kV	128.7	352	26.4
Chancellor Row to Shack Mills Station 138 kV	Chancellor Row Substation to Bear Island 345 kV line	128.2	427	7.8
Muscoda Industrial to Bear Island 138 kV	Chancellor Row to Bear Island 345 kV	125.8	382	7.7
Harrogate to Toro Negro 2 9901 138 kV	Centralia to King City Substation 138 kV	125.3	244	40.8
Shokoe to Moonstone 138 kV	Pheasant Branch to Switzer 138 kV Line	123.6	237	17.9
Basin to Plaza 138 kV	Plaza to Big Hanaford 138 kV	122.5	244	6.0
Northeast to Harrogate 138 kV L	Harrogate to Toro Negro 2 9901 138 kV	121.1	245	62.7
Toro Negro 2 9901 to Anoka Substation 138 kV	Centralia to King City 138 kV L	115.8	234	40.8
Four Rivers to Dps Ocean Spray 138 kV	Leavitt to Birchwood Substation 345 kV	114.8	466	5.5
Four Rvers to Trenton Diesel 138 kV	Leavitt Substation to Birchwood 345 kV	114.8	466	5.5

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bairoa 3008 to Bank Of America 138 kV Line	Fritch to Big Hanaford 138 kV L	113.0	315	11.7
King City to Centralia 138 kV	Fritch Substation to Big Hanaford 138 kV	110.9	497	47.5
Cedar Bayou Plant to Centralia 138 kV	Pheasant Branch to Switzer 138 kV line	107.8	303	24.9
Cedar Bayou Plant to Jim Hill 138 kV	Switzer to Pheasant Branch 138 kV	106.8	352	24.9
Natchez to Lakeside 138 kV	Skaggs Island Jct 2 to Mitchell Dam 345 kV	106.8	625	18.5
Bear to Fluvanna P.S Substation 345 kV L	Leavitt to Birchwood 345 kV	106.3	2,071	14.7
Jim Hill to Haggin 138 kV	Pheasant Brach to Switzer 138 kV	105.9	413	24.9
Fritch to Lakeside 138 kV	Mitchell Dam to Skaggs Island Junction 2 345 kV	104.5	706	18.5
N Randolph to Bairoa 3008 Substation 138 kV L	Fritch to Big Hanaford 138 kV	103.5	323	11.7
Mitchell Dam to Skaggs Island Jct 2 345 kV	Mecklenburg Power Sta to Bear 345 kV	103.5	2,032	21.3
Elko Substation to Syl Laskin 138 kV	Michell Dam to Haggin 345 kV	103.1	378	8.6
North Randolph to Yellow Creek 138 kV	Fritch to Big Hanaford Substation 138 kV	102.7	339	11.7
Fritch to Big Hanaford 138 kV	Mitchell Dam to Skaggs Island Jct 2 Substation 345 kV	102.5	1,017	33.3
Westcoast Cellufibre to Chipola 138 kV L	Skaggs Island Junction 2 to Mitchell Dam 345 kV L	102.4	520	17.6
Strathmore to Chipola 138 kV Line	Fritch to Lakeside 138 kV	102.3	365	21.9
Strathmore to Humacao 2601 138 kV	Fritch to Lakeside 138 kV Line	101.0	398	21.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Muscoda Industrial to Elmont 138 kV	Rebuild line with higher-capacity conductor	\$4,977,000	5% (\$250,000)
Potosi Substation to Deltaport 138 kV line	Rebuild line with higher-capacity conductor	\$5,152,000	10% (\$497,000)
Kimberley to Elmont 138 kV Line	Rebuild line with higher-capacity conductor	\$1,863,000	5% (\$94,000)
Potosi to Tyler 138 kV	Rebuild line with higher-capacity conductor	\$567,000	10% (\$55,000)
Kimberley to Shack Mills Sta 138 kV	Rebuild line with higher-capacity conductor	\$2,972,000	5% (\$149,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Deltaport to Moonstone 138 kV	Rebuild line with higher-capacity conductor	\$2,692,000	10% (\$259,000)
Chancellor Row to Shack Mills Sta 138 kV	Rebuild line with higher-capacity conductor	\$2,544,000	5% (\$128,000)
Muscoda Industrial to Bear Island 138 kV	Rebuild line with higher-capacity conductor	\$8,682,000	5% (\$436,000)
Harrogate to Toro Negro 2 9901 138 kV	Rebuild line with higher-capacity conductor	\$3,512,000	24% (\$838,000)
Moonstone to Shockoe 138 kV	Rebuild line with higher-capacity conductor	\$14,980,000	12% (\$1,874,000)
Basin to Plaza 138 kV	Rebuild line with higher-capacity conductor	\$3,095,000	3% (\$91,000)
Northeast to Harrogate 138 kV L	Rebuild line with higher-capacity conductor	\$2,300,000	27% (\$620,000)
Toro Negro 2 9901 to Anoka 138 kV L	Rebuild line with higher-capacity conductor	\$1,620,000	24% (\$386,000)
Four Rivers Substation to Dps Ocean Spray 138 kV	Rebuild line with higher-capacity conductor	\$3,523,000	5% (\$176,000)
Trenton Diesel to Four Rivers 138 kV	Rebuild line with higher-capacity conductor	\$8,089,000	5% (\$404,000)
Bairoa 3008 to Bank Of America 138 kV Line	Rebuild line with higher-capacity conductor	\$2,078,000	5% (\$101,000)
King City to Centralia 138 kV	Rebuild line with higher-capacity conductor	\$12,036,000	11% (\$1,367,000)
Cedar Bayou Plant to Centralia 138 kV	Reconductor line	\$393,000	8% (\$33,000)
Cedar Bayou Plant to Jim Hill 138 kV	Reconductor line	\$2,220,000	8% (\$184,000)
Natchez to Lakeside 138 kV	Reconductor line	\$1,031,000	4% (\$37,000)
Bear to Fluvanna P.S Substation 345 kV L	Reconductor line	\$42,227,000	1% (\$414,000)
Jim Hill to Haggin 138 kV	Reconductor line	\$337,000	8% (\$28,000)
Lakeide to Fritch 138 kV	Reconductor line	\$1,061,000	4% (\$38,000)
N Randolph to Bairoa 3008 Substation 138 kV L	Reconductor line	\$3,552,000	5% (\$180,000)
Skaggs Island Jct 2 Substation to Mitchell Dam 345 kV	Reconductor line	\$21,338,000	1% (\$271,000)
Elko Substation to Syl Laskin 138 kV	Reconductor line	\$4,889,000	2% (\$112,000)
North Randolph to Yellow Creek 138 kV	Reconductor line	\$2,885,000	5% (\$146,000)
Fritch to Big Hanaford 138 kV	Reconductor line	\$1,728,000	4% (\$72,000)
Chipola to Westcoast Cellufibre 138 kV	Reconductor line	\$1,318,000	3% (\$36,000)
Chipola to Strathmore 138 kV	Reconductor line	\$807,000	7% (\$59,000)
Strathmore to Humacao 2601 138 kV	Reconductor line	\$1,555,000	7% (\$113,000)
	Total allocated to project		\$9,448,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.